the joint surface is covered by a double-layered membrane called the **periosteum**. The outer fibrous layer is dense irregular connective tissue. The inner osteogenic layer consists of **osteoblasts** (bone- forming cells) which secrete bone matrix elements and **osteoclasts** (bone-destroying cells). In addition, there are primitive stem cells, osteogenic cells, that give rise to the osteoblasts. The periosteum is richly supplied with nerve fibres, lymphatic vessels and blood vessels. Internal bone surfaces are covered with a delicate connective tissue membrane called the **endosteum**. The endosteum covers the trabeculae of spongy bone and lines the canals that pass through the compact bone. It also contains both osteoblasts and osteoclasts. Between the epiphysis and diaphysis **epiphyseal plate** or **growth plate** is present.

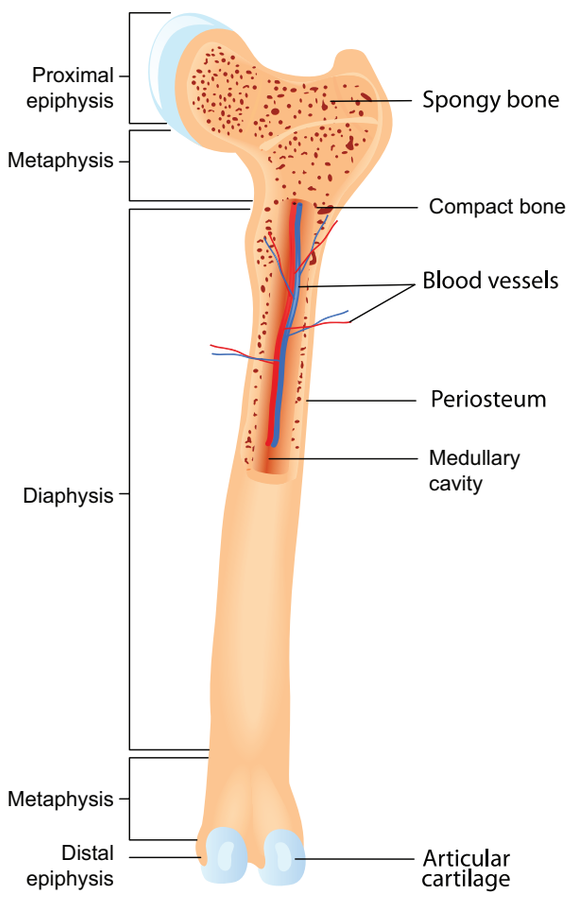


Figure 9.10 Structure of a long bone

9.11 Types of joints

Joints are essential for all types of movements performed by the bony parts of the body. The joints are points of contact (Figure 9.11) between bones.

Sometimes they are playing a protective role in the process. Force generated by the muscles are used to carry out the movement through joints which helps human functional activity of daily living and ambulation. The joint acts as a fulcrum of a lever.

(i) **Fibrous joints or Synarthroses:** They are immovable fixed joints in which no movement between the bones is possible. Sutures of the flat skull bones are fibrous joints.

(ii) **Cartilaginous joints or Amphiarthroses:** They are slightly movable joints in which the joint surfaces are separated by a cartilage and slight movement is only possible. E.g., Joints of adjacent vertebrae of the vertebral column.

(iii) Synovial joints or Diarthroses joints: They are freely movable joints, the articulating bones are seperated by a cavity which is filled with synovial fluid.

Pivot joint	between atlas and axis
Plane/gliding joint	between the carpals
Saddle joint	between the carpal and metacarpal
Ball and socket joint	between humerus and pectoral girdle
Hinge joint	knee joint
Condyloid or Angular or Ellipsoid	between radius and carpal

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